Venkata Lakshmi Parimala Pasupuleti

AI Platform Engineer · Agentic Systems & LLM Evaluation · Internal Developer Tools · RAG Pipelines · PropTech Domain · Python · TypeScript · AWS · GCP · 9yr Production

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PROFESSIONAL SUMMARY

AI platform engineer with 9+ years of production software experience - specializing in building AI evaluation frameworks, agentic workflows, and reusable AI tooling that engineering teams can adopt and trust. AgentProbe is a production-grade custom evaluation harness: CI-integrated regression gate, LLM-as-Judge with pass@k multi-trial sampling, 5-metric quality engine - exactly the internal AI evaluation and governance infrastructure RealPage's CoE is building. LeaseI is the production agentic counterpart: an 8-agent LangGraph pipeline automating compliance investigation in a regulated PropTech domain, with 93.6% precision across 1,240 test cases. At Entrata (7yr, PropTech SaaS serving the same operators as RealPage), I built internal debugging tools that cut triage time 40%, automated 80% of manual operational workflows, and mentored 5+ engineers - the enablement dimension of this role is where I've spent most of my career. At George Mason I mentored 600+ students in Python, SQL, and distributed systems. I work in AI-assisted development daily (Claude Code), with deep hands-on experience in LangGraph, RAG architectures (TF-IDF + ChromaDB), vector databases, prompt engineering, HITL guardrails, and multi-agent orchestration. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI & Agentic Systems: LangGraph, LangChain, Multi-Agent Orchestration, Agentic Workflows, Multi-Step Workflow Design, Tool Calling / Function Calling, LLM Integration, Generative AI, Prompt Engineering, Context Engineering, Few-Shot Prompting, Chain-of-Thought Prompting, Output Validation, HITL / Human-in-the-Loop, Guardrails, RAG Pipelines, Knowledge Management & Retrieval, Semantic Retrieval, Embeddings, Vector Databases (ChromaDB), TF-IDF Retrieval, Document Intelligence, OCR, Confidence Scoring, Anthropic API, OpenAI API, Groq, Ollama

LLM Evaluation & Observability: Custom Evaluation Harnesses, LLM-as-Judge, pass@k Multi-Trial Sampling, Hallucination Detection, Groundedness Scoring, Latency & Token Efficiency Metrics, CI-Integrated Regression Gates, Agent Observability, Distributed Tracing, Structured Logging, Monitoring, Alerting, Feedback Loops, AI Coding Tools (Claude Code, AI-assisted development), Playwright

Languages, Frameworks & APIs: Python (9yr), TypeScript, JavaScript, React.js, FastAPI, Flask, REST APIs, API Design, GraphQL-adjacent, Async Processing, RabbitMQ, Microservices, Distributed Systems, Performance Optimization, Query Optimization, PostgreSQL, MySQL, Oracle, NoSQL, SQLite, Redis

Cloud, DevOps & Testing: AWS (S3, EC2, RDS, Glue, Lambda), Google Cloud Storage (GCP), Docker, CI/CD (GitHub Actions), Automated Testing, Regression Testing, Integration Testing, Unit Testing, Git, Linux, Agile/Scrum, Technical Mentoring, Code Reviews, Design Documents

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - 0→1 Agentic AI Delivery

- Sole technical owner of a production AI system built 0→1: designed agentic workflow architecture (LangGraph stateful graph), built RAG retrieval layer with compliance guardrails, implemented evaluation pipeline and CI validation gate, instrumented observability stack - delivered in weeks with measurable outcomes driving iteration decisions.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate - Engineering Enablement

- Mentored 600+ students in Python, SQL, distributed systems, and data engineering; built multi-source ML pipelines (PyTorch, scikit-learn, R^2=0.977) on AWS and GCP across 190+ countries; applied Isolation Forest anomaly detection across 3 research pipelines; designed Tableau/ggplot2 dashboards improving research comprehension 40%.

Entrata - Multifamily Real Estate SaaS (direct RealPage domain)

Sep 2018 - Aug 2024

Senior Software Engineer - Internal Tooling, Enablement & Production Systems · Lehi, UT (Remote)

- **Internal tooling & engineering enablement** - built debugging and investigation tools reducing team triage time 40%, keeping bug count under 10 for 6+ consecutive months; Python/Bash automation eliminating 80% of manual operational workload; structured logging, monitoring, and alerting for production pipeline oversight; mentored 5+ engineers; 300+ peer code reviews across AI-adjacent data and compliance workflows.
- **Regulated data pipelines at production scale** - owned annual HUD income-limit import pipeline for 7 years: few thousand federal source rows → 180,000 domain records powering eligibility, rent, HAP, and tax-credit calculations for 100K+ housing units; baseline validation, reconciliation, and rollback procedures on every release; zero compliance violations across full ownership period. Direct PropTech domain analog to RealPage's core platform.
- **Full-stack production delivery** - shipped 20+ production modules (Python, JavaScript, TypeScript, React.js, PostgreSQL) across 120+ Agile sprints; RabbitMQ async distributed processing with retry logic and idempotency guarantees; 200+ SQL query optimizations (30% performance gain); Docker microservices redesign (50% scalability improvement); GitHub Actions CI/CD with comprehensive automated test suites.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built property management modules (Python, PostgreSQL); data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints; 56 state-specific document templates.

PROJECTS

AgentProbe: AI Evaluation Framework & Developer Platform - Direct CoE Analog

LLM Evaluation · Custom Harness · LangGraph · CI Gate · Reusable Toolkit · Open Source

Python · FastAPI · LangGraph · React.js · TypeScript · ChromaDB · GitHub Actions · Anthropic API · OpenAI API · Playwright | github.com/venkatalakshmiparimala/AgentProbe

Production AI evaluation toolkit - a reusable developer platform that makes AI systems trustworthy before they reach production; the exact type of internal tooling RealPage's CoE is building. **Evaluation infrastructure:** 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency), LLM-as-Judge with pass@k multi-trial harness detecting output variance, CI-integrated regression gate (GitHub Actions) blocking quality degradation. **Agentic & RAG layer:** LangGraph 3-agent orchestration, ChromaDB semantic retrieval pipeline, embedding-based indexing with structured observability and distributed tracing across all agent steps. **Engineering adoption:** 15-endpoint FastAPI backend, React.js + TypeScript frontend - designed as a reusable, multi-provider toolkit (Anthropic, OpenAI, Groq, Ollama) with zero vendor lock-in. Used to evaluate LeaseI across 1,240+ compliance test cases with zero regressions.

LeaseI: Production Agentic AI Platform - PropTech Compliance Domain

Agentic · LangGraph · RAG · HITL · Regulated Domain · PropTech · Production

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · SQLite · MCP · Anthropic API | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production agentic AI platform for regulated compliance - a working proof that agentic workflows can move from experimentation to repeatable production use in a real domain. **Multi-step agentic workflow:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) with conditional routing, real HITL escalation on low-confidence paths, and deterministic safety boundaries. **RAG & knowledge retrieval:** hand-rolled TF-IDF retrieval over regulatory corpus (section-aware chunking, OCR fallback, 100% citation correctness). **Responsible AI:** LLM reasons and extracts; deterministic logic or human oversight makes every final regulated decision - guardrail enforced in code. **Evaluation:** 93.6% precision, 87.9% F1 across 1,240 test cases; regression gate via AgentProbe CI pipeline.

AI on FHIR: Cloud-Native AI Application - Regulated Healthcare Domain

Full Stack · AWS Lambda · GCP · HIPAA · LLM · RAG · Observability

Python · Flask · React.js · TypeScript · Docker · AWS Lambda · GCP · PostgreSQL · REST APIs · spaCy · BERT | github.com/venkatalakshmiparimala/AI_on_FHIR

Cloud-native AI application for a privacy-regulated domain - full-stack LLM platform translating natural-language queries (EN/ES) into structured FHIR API calls; HIPAA-compliant RBAC with complete audit trail for regulated-data access; AWS Lambda + GCP backend services; Docker containerization; observability stack with request logging and compliance tracing. Demonstrates AI solution development in environments where privacy, compliance, and sensitive-data handling are first-class requirements - ~60% reduction in manual clinical retrieval time.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017